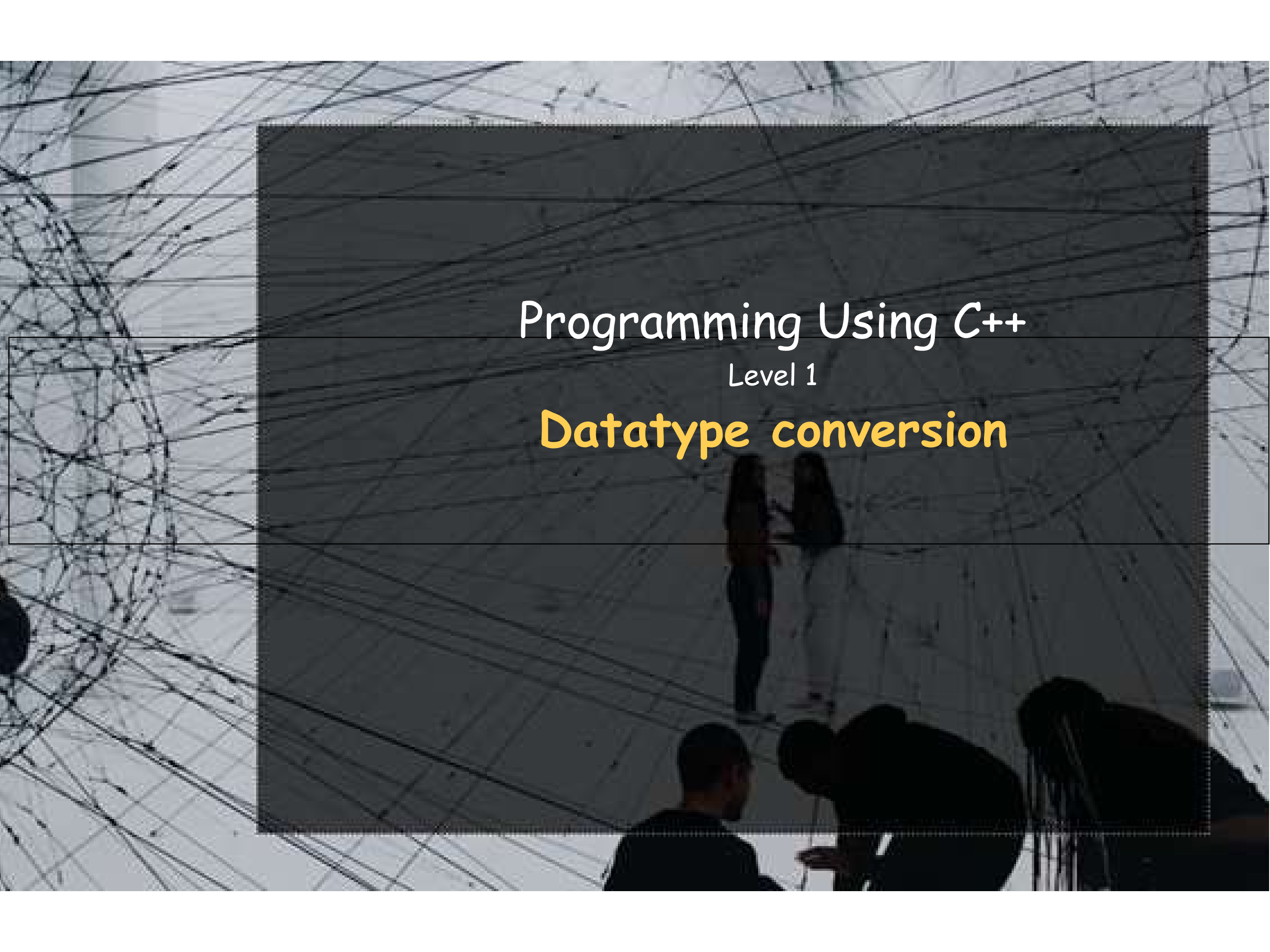


Programming Using C++ Level 1

Datatype conversion



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Definition

◆ Datatype conversion is the process of changing a value from one data type to another.

◆ In this program, C++ converts:

▣ `string` → `double`

▣ `string` → `float`

▣ `string` → `int`

▣ `int` → `string`

▣ `double` → `string`

▣ `float` → `string`

Explanation

◆ The program demonstrates how to convert between `string` and numerical data types using functions from the C++ standard library.

◆ The functions used are:

- ▣ `stod()` → converts a string to `double`.

- ▣ `stof()` → converts a string to `float`.

- ▣ `stoi()` → converts a string to `int`.

- ▣ `to\_string()` → converts a numerical value to `string`.

Steps in main()

1 Create a string

- ◆ The program creates a string containing the numerical value `"43.22".

2 Convert string to double

- ◆ `stod()` converts the string into a `double`.

3 Convert string to float

- ◆ `stof()` converts the string into a `float`.

4 Convert string to int

- ◆ `stoi()` converts the string into an `int`.

5 Convert int to string

- ◆ `to\_string()` converts the integer `20` into a string.

6 Convert double to string

- ◆ `to\_string()` converts the double `33.5` into a string.

7 Convert float to string

- ◆ `to\_string()` converts the float `55.23` into a string.

💡 Example

◆ Given:

```
string St1 = "43.22";
```

◆ The conversions produce:

```
stod(St1) → 43.22
```

```
stof(St1) → 43.22
```

```
stoi(St1) → 43
```

➡ Result

◆ `stod()` and `stof()` keep the decimal value, while `stoi()` converts the value to an integer and removes the decimal part.

Note

- ◆ `stoi()` converts the integer part of a numeric string.

- ◆ For example:

 - `stoi("43.22")`

- ◆ produces:

 - 43

⊘ Warning

- ◆ The string must contain a valid numerical value that the conversion function can interpret.
- ◆ Trying to convert invalid text, such as ``"Hello"`,` to a number can cause a conversion error.



Comparison

▨ ``stod()`, `stof()`, `stoi()``

- ✓ Convert a ``string`` into a numerical data type.
- ✓ Used when numerical calculations are needed.
- ✓ Example: ``stoi("20")`` → ``20``

▤ ``to_string()``

- ✓ Converts a numerical value into a ``string``.
- ✓ Useful when a number needs to be treated as text.
- ✓ Example: ``to_string(20)`` → ``"20"``

✦✦ Summary

- ✓ Datatype conversion changes a value from one data type to another.
- ✓ ``stod()`, `stof()`, and `stoi()`` convert strings into numerical values.
- ✓ ``to_string()`` converts numerical values into strings.

Thank you 😊